



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO RAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

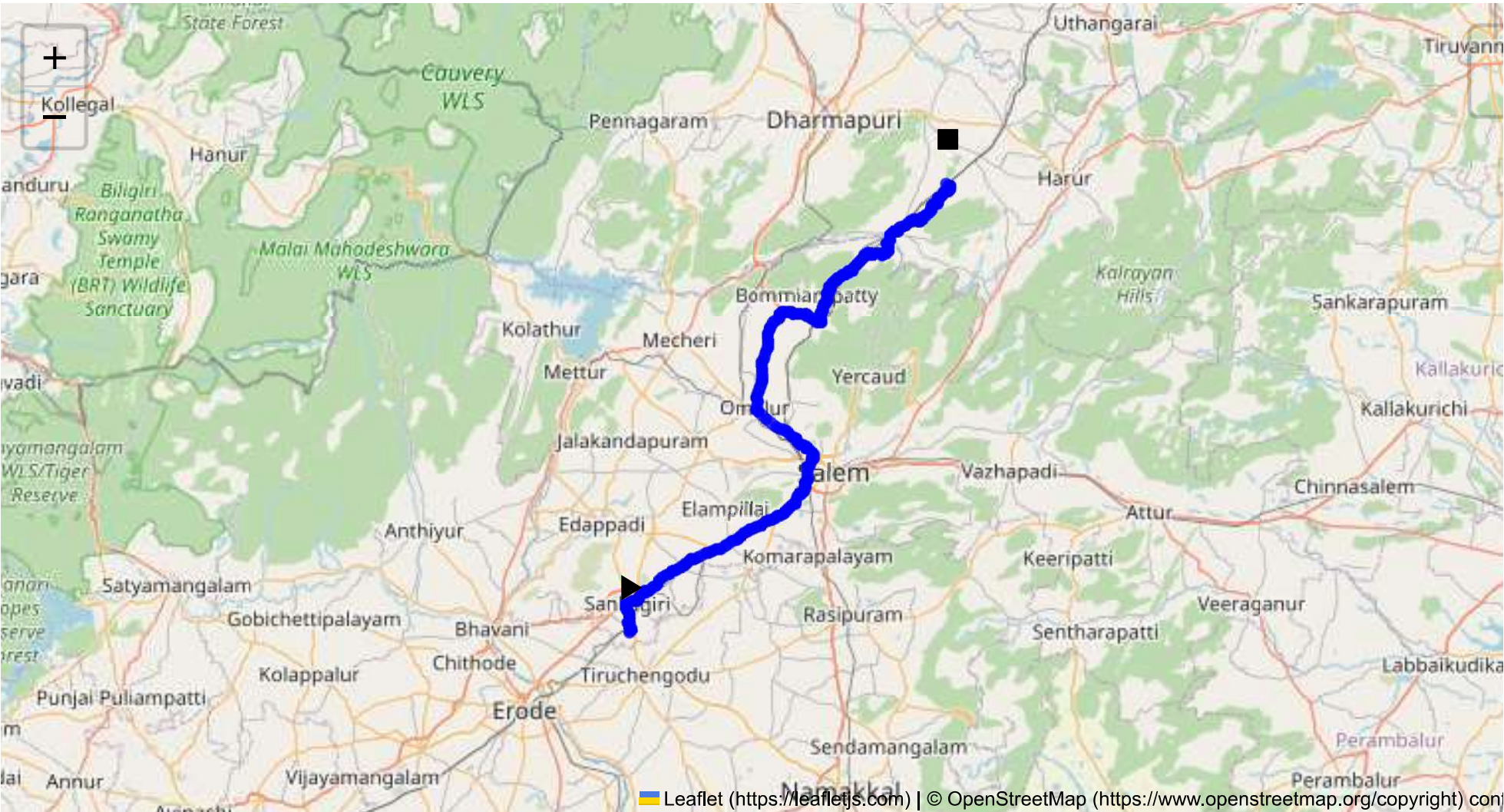
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 110.31 km
Estimated Duration: 2.5 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours
Start: (11.4381, 77.8734)
End: (12.048268, 78.312705)



Welcome to the Journey Risk Management Study

To provide a comprehensive analysis of the route from CVQF+23W, Sangagiri, Tamil Nadu to 28W7+W9M, Tamil Nadu, and ensure the safety of truck drivers transporting hazardous materials, I have compiled detailed findings across the specified points. Please note that this analysis leverages historical data and general insights, but real-time monitoring and logistics analysis will be crucial for current conditions.

1. Overview of the Route Map:

- The route covers a stretch of approximately 110.31 kilometers, traversing rural and semi-urban sectors in Tamil Nadu. The journey progresses primarily along NH-44 with diversions via state highways as needed.

2. Typical Weather Conditions and Potential Weather-related Hazards:

- The climate in Tamil Nadu is generally tropical with heavy monsoon rains from June to September. During these months, the roads can be slippery, with poor visibility and potential flooding in low-lying areas causing hazardous driving conditions.

3. Analysis of Traffic Patterns:

- Urban areas along NH-44 such as Salem and points around populated towns often experience congestion. Peak hours are typically 8-10 AM and 6-8 PM. Major junctions and intersections are more prone to delays.

4. Assessment of Road Quality and Infrastructure:

- The national highway segment is generally well-maintained. However, state roads may vary in quality, with possible potholes and narrower lanes closer to villages. Road signage may be sparse in rural areas, necessitating caution and GPS support.

5. Suggestions for Alternative Routes for Emergencies:

- Alternative routes could involve bypassing congested town centers via less populated routes like rural link roads. Emergency bypasses should avoid flood-prone areas during the monsoon.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Transport of hazardous materials mandates strict adherence to government permits and guidelines, especially regarding dangerous goods. Night transport may be restricted, and dedicated safety equipment is required on board.

7. Overview of Historical Incidents:

- Historical data shows previous incidents related to fog, over-speeding, and technical failures leading to accidents. No major incidents specific to hazardous materials have been reported recently but general caution is reinforced.

8. Environmental Considerations and Sensitive Areas:

- The route may traverse regions with dense vegetation and wildlife. Care should be taken to avoid contamination in agricultural zones and sensitive ecosystems.

9. Analysis of Communication Coverage:

- Mobile coverage is generally reliable along the NH-44 corridor, with potential dead zones as you move onto state managed roads in rural areas, particularly in more remote stretches.

10. Estimated Emergency Response Times:

- Emergency services in urban centers like Salem may take 20-30 minutes to deploy. Rural areas could experience delays of up to an hour due to distance and road conditions.

11. Overall Summary of Risk Assessment:

- While the major part of the route along NH-44 is secure and well-facilitated, potential risks include inclement weather, peak-hour congestion, inconsistent road quality in rural sections, and

communication dead zones. Effective route planning, adherence to safety guidelines, and real-time navigation updates can mitigate most risks.

Key recommendations for drivers include pre-trip inspections, avoiding travel during peak monsoon periods, following regulated speed limits, maintaining updated communication devices, and possessing precise documentation for materials transported.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.45309, 77.87325	30 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	Medium	11.46765, 77.87037	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.53501, 77.95707	30 KM/Hr
11	Turn	Medium	11.53601, 77.95856	30 KM/Hr
12	Turn	Medium	11.62403, 78.10893	30 KM/Hr
13	Turn	Medium	11.64180, 78.11808	30 KM/Hr
14	Turn	Medium	11.64325, 78.11993	30 KM/Hr
15	Turn	Medium	11.69519, 78.10775	30 KM/Hr
16	Turn	Medium	11.73958, 78.05105	30 KM/Hr
17	Turn	Medium	11.75835, 78.05016	30 KM/Hr
18	Turn	Medium	11.86847, 78.08367	30 KM/Hr
19	Turn	High	11.87086, 78.08362	15 KM/Hr
20	Turn	High	11.87111, 78.08382	15 KM/Hr
21	Turn	Medium	11.87070, 78.08484	30 KM/Hr
22	Turn	Medium	11.87098, 78.08571	30 KM/Hr
23	Turn	High	11.87255, 78.08723	15 KM/Hr
24	Turn	Medium	11.86873, 78.12120	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
25	Turn	Medium	11.86484, 78.12547	30 KM/Hr
26	Turn	High	11.85834, 78.13562	15 KM/Hr
27	Turn	Medium	11.86054, 78.13948	30 KM/Hr
28	Turn	Medium	11.86323, 78.14024	30 KM/Hr
29	Turn	Medium	11.86644, 78.13881	30 KM/Hr
30	Turn	Medium	11.88100, 78.14378	30 KM/Hr
31	Turn	Medium	11.88387, 78.14463	30 KM/Hr
32	Turn	Medium	11.90592, 78.15290	30 KM/Hr
33	Turn	High	11.90775, 78.15206	15 KM/Hr
34	Turn	Medium	11.91500, 78.15813	30 KM/Hr
35	Turn	High	11.92457, 78.17082	15 KM/Hr
36	Turn	Medium	11.92428, 78.17192	30 KM/Hr
37	Turn	Medium	11.92810, 78.18001	30 KM/Hr
38	Turn	Medium	11.92795, 78.18186	30 KM/Hr
39	Turn	Medium	11.92907, 78.18352	30 KM/Hr
40	Turn	Medium	11.93578, 78.18719	30 KM/Hr
41	Turn	Medium	11.94936, 78.20204	30 KM/Hr
42	Turn	Medium	11.95139, 78.20667	30 KM/Hr
43	Turn	Medium	11.95216, 78.20724	30 KM/Hr
44	Turn	High	11.95255, 78.20816	15 KM/Hr
45	Turn	Medium	11.95175, 78.20993	30 KM/Hr
46	Turn	Medium	11.95213, 78.21201	30 KM/Hr
47	Turn	High	11.95159, 78.21293	15 KM/Hr
48	Turn	Medium	11.95236, 78.21435	30 KM/Hr
49	Turn	High	11.95221, 78.21499	15 KM/Hr
50	Turn	Medium	11.95266, 78.21563	30 KM/Hr
51	Turn	Medium	11.95259, 78.22493	30 KM/Hr
52	Turn	High	11.95179, 78.22608	15 KM/Hr
53	Turn	Medium	11.96149, 78.23446	30 KM/Hr
54	Turn	Medium	11.96527, 78.23288	30 KM/Hr
55	Turn	High	11.96634, 78.23303	15 KM/Hr
56	Blind Spot	Blind Spot	11.96687, 78.23203	10 KM/Hr
57	Turn	High	11.97325, 78.23714	15 KM/Hr
58	Turn	High	11.97747, 78.23549	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
59	Turn	High	11.98245, 78.24301	15 KM/Hr
60	Turn	High	11.98222, 78.24321	15 KM/Hr
61	Turn	High	11.99210, 78.25772	15 KM/Hr
62	Turn	Medium	11.99188, 78.25802	30 KM/Hr
63	Turn	High	11.99183, 78.25890	15 KM/Hr
64	Turn	Medium	11.99287, 78.26006	30 KM/Hr
65	Turn	Medium	11.99367, 78.26036	30 KM/Hr
66	Turn	Medium	11.99939, 78.27157	30 KM/Hr
67	Turn	Medium	11.99924, 78.27272	30 KM/Hr
68	Turn	High	11.99674, 78.27753	15 KM/Hr
69	Turn	High	11.99812, 78.27825	15 KM/Hr
70	Turn	High	11.99800, 78.27978	15 KM/Hr
71	Turn	Medium	11.99835, 78.28018	30 KM/Hr
72	Turn	Medium	12.00120, 78.28087	30 KM/Hr
73	Turn	Medium	12.01914, 78.29621	30 KM/Hr
74	Turn	Medium	12.02046, 78.30108	30 KM/Hr
75	Turn	Medium	12.02240, 78.30274	30 KM/Hr
76	Turn	Medium	12.02850, 78.30342	30 KM/Hr
77	Turn	High	12.02936, 78.30415	15 KM/Hr
78	Turn	High	12.02905, 78.30515	15 KM/Hr
79	Turn	High	12.03610, 78.31161	15 KM/Hr
80	Turn	Medium	12.03586, 78.31346	30 KM/Hr
81	Turn	High	12.04334, 78.31907	15 KM/Hr
82	Turn	Medium	12.04565, 78.31487	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
20	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
21	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
22	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
24	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
25	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
26	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
27	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
28	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
31	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
32	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
33	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
34	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
35	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
36	clinic	Kamala Clinic	11.8723673, 78.0874195	30 km/h	Medium
37	hospital	Kamala Hospital	11.8722588, 78.0899189	30 km/h	Medium
38	hospital	GJB Hospital	11.8713998, 78.0980288	30 km/h	Medium
39	clinic	Pranav Clinic	11.8712846, 78.0998167	30 km/h	Medium
40	hospital	Bethel Hospital	11.8699472, 78.1388598	30 km/h	Medium
41	hospital	Amirtham Hospital	11.981994, 78.248038	30 km/h	Medium
42	hospital	Grs Hospital	11.9824068, 78.2481763	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
29	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
30	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64325, 78.11993



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69519, 78.10775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73958, 78.05105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87086, 78.08362



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87111, 78.08382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87070, 78.08484



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87098, 78.08571



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87255, 78.08723



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86873, 78.12120



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97325, 78.23714



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97747, 78.23549



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98245, 78.24301



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98222, 78.24321



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.99210, 78.25772



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99188, 78.25802



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.99183, 78.25890



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99287, 78.26006



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99367, 78.26036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99939, 78.27157



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99924, 78.27272



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.99674, 78.27753



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.99812, 78.27825



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.99800, 78.27978



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99835, 78.28018



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.00120, 78.28087



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01914, 78.29621



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.02046, 78.30108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.02240, 78.30274



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.02850, 78.30342



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.02936, 78.30415



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.02905, 78.30515



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.03610, 78.31161

